

Problem 2(a) — Full Walkthrough

Log-sum-exp is convex

Goal: Show that

$$f(x) = \log \left(e^{a_1^T x + b_1} + e^{a_2^T x + b_2} + \dots + e^{a_n^T x + b_n} \right)$$

is convex, where each a_i is a vector and b_i is a scalar.

Strategy: Split this into an “outer” log-sum-exp function and an “inner” affine map. Show the outer piece is convex on its own, then argue that plugging affine expressions into a convex function preserves convexity.

Step 1 — Recognize the structure

Each $a_i^T x + b_i$ is an affine function of x . Stack them: let A be the matrix whose rows are $a_1^T, a_2^T, \dots, a_n^T$, and $b = (b_1, \dots, b_n)$. Then

$$Ax + b = (a_1^T x + b_1, a_2^T x + b_2, \dots, a_n^T x + b_n).$$

Define g , taking any vector $z = (z_1, \dots, z_n)$:

$$g(z) = \log(e^{z_1} + e^{z_2} + \dots + e^{z_n}) \quad (\text{the } \log\text{-sum-exp function}).$$

With this, $f(x) = g(Ax + b)$: apply the affine map $x \mapsto Ax + b$, then feed the result into g .

Step 2 — Why “convex g , affine input” gives convex f

Suppose g is convex. Take any x, y and $\theta \in [0, 1]$. Let $u = Ax + b$, $v = Ay + b$ — two full vectors, not coordinates of one vector.

Because $Ax + b$ is affine, it distributes over convex combinations:

$$\begin{aligned} A(\theta x + (1 - \theta)y) + b &= \theta(Ax) + (1 - \theta)(Ay) + b && [\text{linearity of } A] \\ &= \theta(Ax) + (1 - \theta)(Ay) + \theta b + (1 - \theta)b && [\text{since } \theta + (1 - \theta) = 1, \text{ split } b] \\ &= \theta(Ax + b) + (1 - \theta)(Ay + b) \\ &= \theta u + (1 - \theta)v. \end{aligned}$$

So

$$f(\theta x + (1 - \theta)y) = g(\theta u + (1 - \theta)v) \leq \theta g(u) + (1 - \theta)g(v) = \theta f(x) + (1 - \theta)f(y),$$

using convexity of g in the middle step, and $g(u) = f(x)$, $g(v) = f(y)$ in the last. That’s exactly the definition of f convex. So: **if g is convex, $f = g(Ax + b)$ is automatically convex too**, regardless of A, b .

Step 3 — Compute the gradient of g

Let $S(z) := e^{z_1} + e^{z_2} + \dots + e^{z_n}$, so $g(z) = \log(S(z))$. By the chain rule (log outer, S inner):

$$\frac{\partial g}{\partial z_i} = \frac{1}{S(z)} \cdot \frac{\partial S}{\partial z_i} = \frac{1}{S(z)} \cdot e^{z_i} = \frac{e^{z_i}}{S(z)}.$$

Define the **softmax**:

$$p_i(z) := \frac{e^{z_i}}{S(z)}, \quad i = 1, \dots, n.$$

Each $p_i(z) > 0$, and they sum to 1:

$$p_1 + p_2 + \dots + p_n = \frac{e^{z_1} + \dots + e^{z_n}}{S(z)} = \frac{S(z)}{S(z)} = 1.$$

So $p(z) = (p_1, \dots, p_n)$ is a probability distribution, and

$$\nabla g(z) = p(z).$$

Step 4 — Compute the Hessian of g

Differentiate $p_i(z) = e^{z_i}/S(z)$ again, via the quotient rule.

Diagonal entries (differentiate p_i w.r.t. z_i again):

$$\frac{\partial p_i}{\partial z_i} = \frac{e^{z_i} S(z) - e^{z_i} \cdot e^{z_i}}{S(z)^2} = \frac{e^{z_i} (S(z) - e^{z_i})}{S(z)^2} = p_i(1 - p_i).$$

Off-diagonal entries (differentiate p_i w.r.t. a different variable z_j , $j \neq i$):

$$\frac{\partial p_i}{\partial z_j} = \frac{0 \cdot S(z) - e^{z_i} \cdot e^{z_j}}{S(z)^2} = -p_i p_j.$$

So the Hessian matrix is:

$$\nabla^2 g(z) = \begin{pmatrix} p_1(1-p_1) & -p_1 p_2 & \dots & -p_1 p_n \\ -p_2 p_1 & p_2(1-p_2) & \dots & -p_2 p_n \\ \vdots & \vdots & \ddots & \vdots \\ -p_n p_1 & -p_n p_2 & \dots & p_n(1-p_n) \end{pmatrix} = \text{diag}(p) - pp^T,$$

where $\text{diag}(p)$ is the diagonal matrix with $p_1, \dots, p_n$ on the diagonal, and pp^T is the outer product of p with itself:

$$\text{diag}(p) = \begin{pmatrix} p_1 & 0 & \dots & 0 \\ 0 & p_2 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & p_n \end{pmatrix}, \quad pp^T = \begin{pmatrix} p_1 p_1 & p_1 p_2 & \dots & p_1 p_n \\ p_2 p_1 & p_2 p_2 & \dots & p_2 p_n \\ \vdots & \vdots & \ddots & \vdots \\ p_n p_1 & p_n p_2 & \dots & p_n p_n \end{pmatrix}.$$

Subtracting: the diagonal of $\text{diag}(p) - pp^T$ is $p_i - p_i^2 = p_i(1 - p_i)$ (matches), and the off-diagonal entries are $0 - p_i p_j = -p_i p_j$ (matches).

Step 5 — Show the Hessian is positive semidefinite

Take any vector $v = (v_1, \dots, v_n)$ (a fresh test vector, unrelated to the $v = Ay + b$ from Step 2). Compute $v^T \nabla^2 g(z) v$:

$$\begin{aligned} v^T \nabla^2 g(z) v &= \sum_i p_i (1 - p_i) v_i^2 - \sum_{i \neq j} p_i p_j v_i v_j \\ &= \sum_i p_i v_i^2 - \sum_i p_i^2 v_i^2 - \sum_{i \neq j} p_i p_j v_i v_j \\ &= \sum_i p_i v_i^2 - \left(\sum_i p_i v_i \right)^2, \end{aligned}$$

since $\sum_i p_i^2 v_i^2 + \sum_{i \neq j} p_i p_j v_i v_j$ is exactly the expansion of $\left(\sum_i p_i v_i \right)^2$.

Since p is a probability distribution, treating $v_1, \dots, v_n$ as values with probabilities $p_1, \dots, p_n$:

$$v^T \nabla^2 g(z) v = \mathbb{E}_p[v^2] - (\mathbb{E}_p[v])^2 = \text{Var}_p(v) \geq 0,$$

since variance is never negative — it's an average of squared deviations from the mean.

So $\nabla^2 g(z) \succeq 0$ for every z : the second-order condition for g to be convex.

Step 6 — Conclude

g is convex (Step 5), and $f(x) = g(Ax + b)$ is a convex function composed with an affine map, which is convex (Step 2). Therefore f is convex. ■